

二重积分的计算公式

$$\iint_D f(x,y) dx dy = \int_a^b \left(\int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y) dy \right) dx = \int_a^b dx \int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y) dy$$

例 6.2.1 计算二重积分 $\iint_D x^2 y dx dy$, D 为 $y = \sqrt{x}$, $x = 1$, $y = 0$ 围成的平面区域

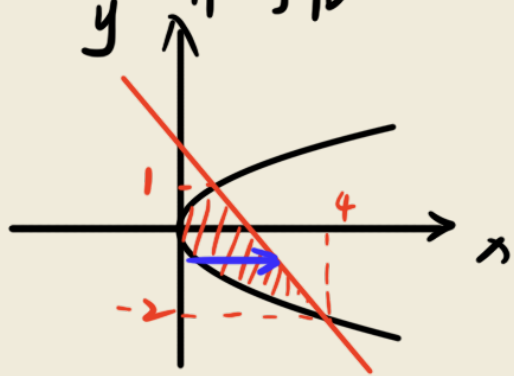
法一: $D = \{(x,y) \mid 0 \leq y \leq \sqrt{x}, 0 \leq x \leq 1\}$.

$$\text{原式} = \int_0^1 dx \int_0^{\sqrt{x}} x^2 y dy = \int_0^1 x^2 \cdot \frac{1}{2} y^2 \Big|_0^{\sqrt{x}} dx = \frac{1}{2} \int_0^1 x^3 dx = \frac{1}{8} x^4 \Big|_0^1 = \frac{1}{8}$$

法二: $D = \{(x,y) \mid y^2 \leq x \leq 1, 0 \leq y \leq 1\}$

$$\text{原式} = \int_0^1 dy \int_{y^2}^1 x^2 y dx = \int_0^1 y \cdot \frac{1}{3} x^3 \Big|_{y^2}^1 dy = \int_0^1 \left(\frac{1}{3} y - \frac{y^7}{3} \right) dy = \left(\frac{y^2}{6} - \frac{y^8}{24} \right) \Big|_0^1 = \frac{1}{8}$$

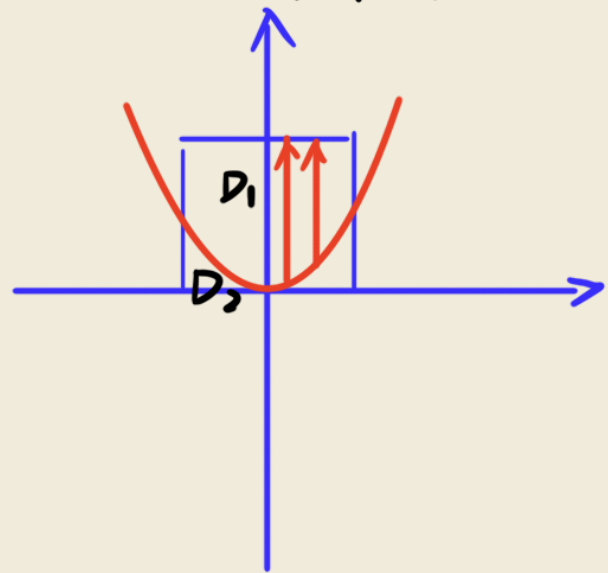
例 6.2.2 计算 $\iint_D (x+y) dx dy$, D 为 $x = y^2$ 及 $x+y=2$ 所围区域.



$$D = \{(x,y) \mid y^2 \leq x \leq 2-y, -2 \leq y \leq 1\}$$

$$\text{原式} = \int_{-2}^1 dy \int_{y^2}^{2-y} (x+y) dx = \int_{-2}^1 \left(\frac{1}{2} x^2 + xy \right) \Big|_{x=y^2}^{x=2-y} dy = \int_{-2}^1 \left(2 - \frac{1}{2} y^2 - \frac{1}{2} y^4 - y^3 \right) dy = \frac{99}{20}$$

例 6.2.3. 计算 $\iint_D |y-x^2| dx dy$ D 为 $|x| \leq 1, 0 \leq y \leq 2$

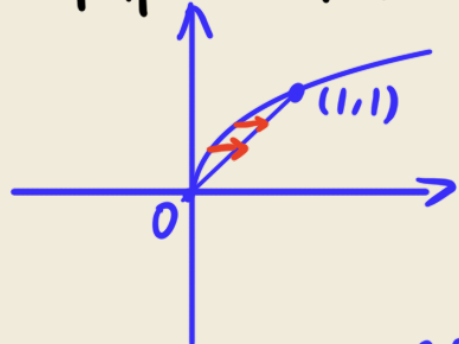


$$= \iint (y-x^2) dx dy + \iint (x^2-y) dx dy$$

$$D_1 = \{(x,y) \mid y \geq x^2\} \quad D_2 = \{(x,y) \mid y \leq x^2\}$$

$$= \int_{-1}^1 dx \int_{x^2}^2 (y-x^2) dy + \int_{-1}^1 dx \int_0^{x^2} (y-x^2) dy$$

例：计算累次积分为 $\int_0^1 dx \int_x^{\sqrt{x}} \frac{\cos y}{y} dy$



找到对应的区域

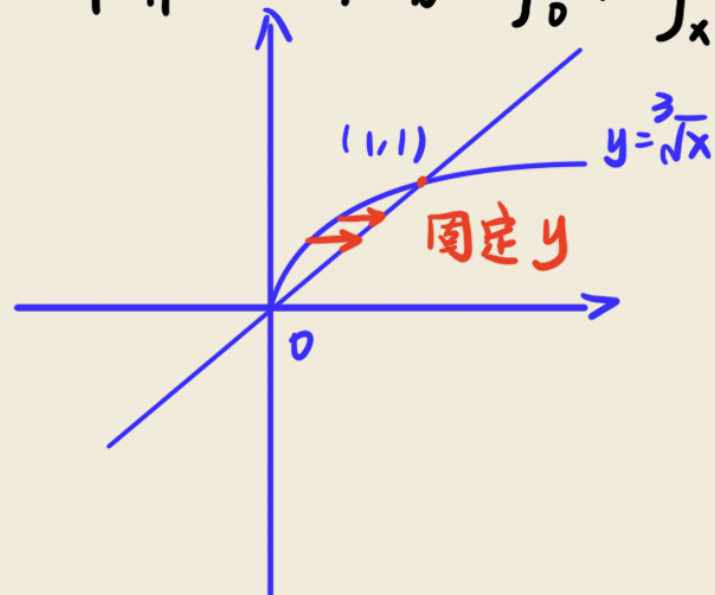
变回重积为

$$\iint_D \frac{\cos y}{y} dx dy$$

$$D = \{(x,y) \mid 0 \leq y \leq 1, y^2 \leq x \leq y\}$$

$$= \int_0^1 dy \int_{y^2}^y \frac{\cos y}{y} dx = \int_0^1 (\cos y - y \cdot \cos y) dy = \sin y \Big|_0^1 - y \sin y \Big|_0^1 - \cos y \Big|_0^1 = 1 - \cos 1$$

例: 计算累次积分 $\int_0^1 dx \int_x^{\sqrt[3]{x}} e^{-y^2} dy$



$$\iint e^{-y^2} dy$$

$$D = \{(x, y) \mid x \leq y \leq \sqrt[3]{x}, 0 \leq x \leq 1\}$$

$$\int_0^1 dy \int_{y^3}^y e^{-y^2} dx = \int_0^1 y \cdot e^{-y^2} - y^3 e^{-y^2} dy$$

= 重积分的积分换元公式

设 D 为平面有界区域. $x = x(u, v), y = y(u, v)$ 在 uv 平面的有界闭区域 D' 连续可微 使 D 与 D' 上的点一一对应 并且有

$$J(u, v) = \frac{D(x, y)}{D(u, v)} \neq 0, (u, v) \in D'$$

换元积分公式: $\iint_D f(x, y) d\sigma = \iint_{D'} f(x(u, v), y(u, v)) |J(u, v)| du dv.$

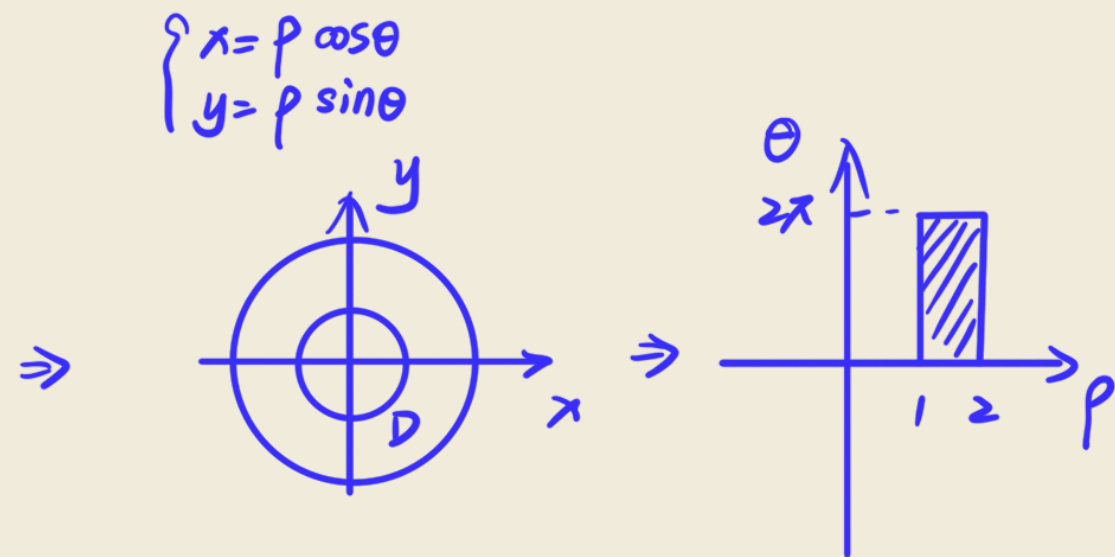
$d\sigma$ 为 $|J(u, v)| du dv$ 是曲线坐标下的面积微元

极坐标变换

$$\begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta \end{cases} \quad J(\rho, \theta) = \frac{D(x, y)}{D(\rho, \theta)} = \begin{vmatrix} \cos \theta & -\rho \sin \theta \\ \sin \theta & \rho \cos \theta \end{vmatrix} = \rho$$

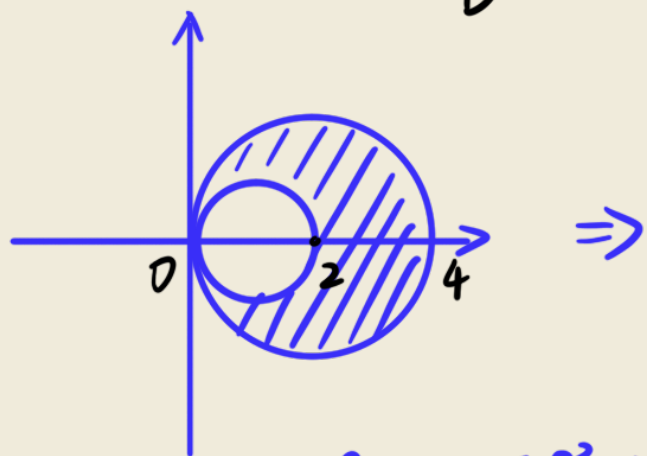
$$\iint_D f(x, y) d\sigma = \iint_{D'} f(\rho \cos \theta, \rho \sin \theta) \rho d\rho d\theta$$

例: 计算 = 重积分 $\iint_D e^{-(x^2+y^2)} dx dy$ D 为区域 $1 \leq x^2+y^2 \leq 4$



故原式 = $\int_1^2 d\rho \int_0^{2\pi} e^{-\rho^2} \rho d\theta = 2\pi \int_1^2 \rho \cdot e^{-\rho^2} d\rho = 2\pi \left(-\frac{1}{2} e^{-\rho^2} \right) \Big|_1^2 = \pi \cdot \left(\frac{1}{e} - \frac{1}{e^4} \right)$

例: 计算 = 重积分 $\iint_D (x+y)^2 dx dy$ D 是由 $2x \leq x^2+y^2 \leq 4x$



$$2\rho \cos \theta \leq \rho^2 \Rightarrow 2 \cos \theta \leq \rho \Rightarrow \cos \theta \leq \frac{\rho}{2}$$

$$\rho^2 \leq 4\rho \cos \theta \Rightarrow \rho \leq 4 \cos \theta \Rightarrow \cos \theta \geq \frac{\rho}{4}$$

$$= \iint_{D'} \rho^3 d\rho d\theta = \int_0^{\frac{\pi}{2}} d\theta \int_{2\cos \theta}^{4\cos \theta} \rho^3 d\rho = \frac{45\pi}{2}$$

例: 计算 $\iint_D (x+y) dx dy$ D 为区域 $x^2+y^2 \leq 2y$

$$\iint_D (x+y) dx dy = 2 \iint_{D_1} y \cdot dx dy$$

$$D' = \{(p, \theta) \mid 0 \leq p \leq 2 \sin \theta, 0 \leq \theta \leq \frac{\pi}{2}\}$$

$$\text{原式} = 2 \iint_{D'} p^2 \sin \theta dp d\theta = 2 \int_0^{\frac{\pi}{2}} d\theta \int_0^{2 \sin \theta} p^2 \sin \theta dp = \pi$$

例: 计算 $\iint_D (1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}) dx dy$, D 为区域 $\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$ ($a, b > 0$)

$$\begin{cases} x = ap \cos \theta \\ y = bp \sin \theta \end{cases}$$

$$J(p, \theta) = \frac{D(x, y)}{D(p, \theta)} = \begin{vmatrix} a \cos \theta & -ap \sin \theta \\ b \sin \theta & bp \cos \theta \end{vmatrix} = abp$$

$$D' = \{(p, \theta) \mid 0 \leq p \leq 1, 0 \leq \theta \leq 2\pi\}$$

$$\text{原式} = \iint_{D'} abp (1 - p^2) dp d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^1 abp (1 - p^2) dp = \frac{ab}{2} \pi$$

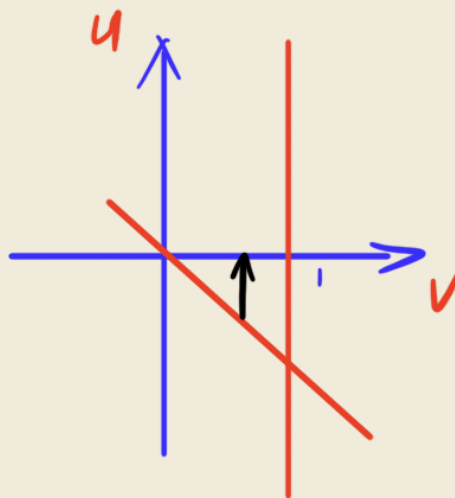
例 计算 $\iint_D e^{\left(\frac{y-x}{y+x}\right)} dx dy$, 区域 D 为 $y=x, y=0, x+y=1$

$$\begin{cases} u = y - x \\ v = y + x \end{cases}$$

$$\Rightarrow \begin{cases} x = \frac{v-u}{2} \\ y = \frac{v+u}{2} \end{cases}$$

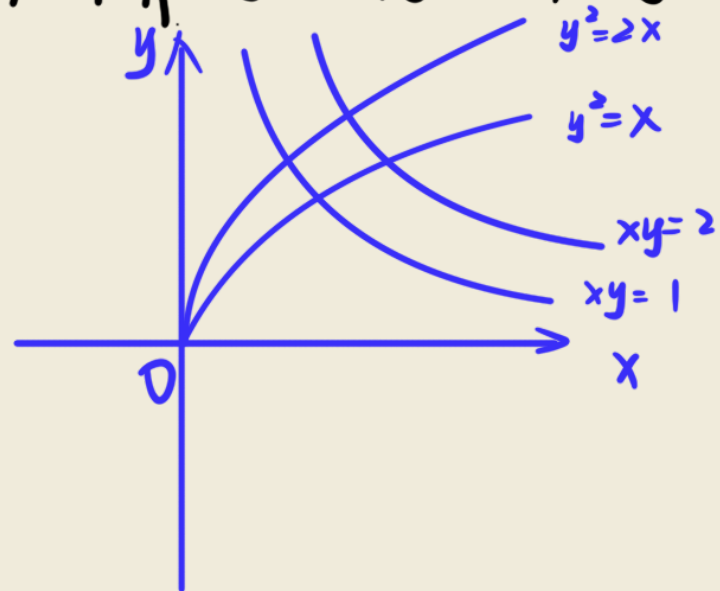
$$J(u,v) = \frac{D(x,y)}{D(u,v)} = \begin{vmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} = -\frac{1}{2}$$

$$D' = \{(u,v) \mid -v \leq u \leq 0, 0 \leq v \leq 1\}$$



$$\begin{aligned} \text{原式} &= \iint_{D'} \frac{1}{2} e^{\frac{u}{v}} du dv \\ &= \frac{1}{2} \int_0^1 dv \int_{-v}^0 e^{\frac{u}{v}} du \end{aligned}$$

例: 计算 $y^2 = x$, $y^2 = 2x$, $xy = 1$, $xy = 2$ 所围平面区域 D 的面积



$$\begin{aligned} u &= \frac{y^2}{x} \\ v &= xy \end{aligned} \Rightarrow \begin{aligned} x &= \sqrt[3]{\frac{v^2}{u}} \\ y &= \sqrt[3]{uv} \end{aligned}$$

$$J(u,v) = \frac{D(x,y)}{D(u,v)} = \begin{vmatrix} -\frac{1}{3} \sqrt[3]{\frac{v^2}{u^4}} & \frac{2}{3} \sqrt[3]{\frac{1}{uv}} \\ \frac{1}{3} \sqrt[3]{\frac{v}{u^2}} & \frac{1}{3} \sqrt[3]{\frac{u}{v^2}} \end{vmatrix} = -\frac{1}{3u}$$

$$\text{而 } \frac{D(x,y)}{D(u,v)} \cdot \frac{D(u,v)}{D(x,y)} = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} \cdot \begin{bmatrix} -\frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

互逆矩阵.

$$J = \left| \frac{D(u,v)}{D(x,y)} \right|^{-1} = \left| \begin{vmatrix} -\frac{y^2}{x^2} & \frac{2y}{x} \\ y & x \end{vmatrix} \right|^{-1} = \left| -\frac{2y^2}{x} \right|^{-1} = \frac{1}{3u} \quad \text{取正!}$$

$$D' = (u, v) = \{(u, v) \mid 1 \leq u \leq 2, 1 \leq v \leq 2\}$$

$$G(D) = \iint_D dx dy = \iint_{D'} \frac{1}{3u} du dv = \int_1^2 dv \int_1^2 \frac{1}{3u} du = \frac{1}{3} \ln 2$$

$$\iint_D x^3 y dx dy \quad D \text{ 是由 } y = \frac{x^3}{2}, y = x^3, x = y^2, x = \frac{y^2}{2} \text{ 所围的区域}$$

$$u = \frac{x^3}{y} \quad (1, 2)$$

$$v = \frac{y^2}{x} \quad (1, 2)$$

$$J(u, v) = \left| \frac{D(u, v)}{D(x, y)} \right| = \begin{vmatrix} \frac{3x^2}{y} & -\frac{x^3}{y^2} \\ -\frac{y^2}{x^2} & \frac{2y}{x} \end{vmatrix} = |5x| = \frac{1}{5x}$$

$$\text{代入 求 } -\frac{1}{5} \iint_D x^2 y dx dy$$

$$\frac{1}{5} \iint_{D'} uv du dv$$

$$\frac{1}{5} \cdot \frac{1}{2} \cdot 3 \cdot \frac{1}{2} \cdot 3 = \frac{9}{20}$$